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(54) **ROOFING SYSTEM INCLUDING
PHOTOVOLTAIC MODULE WIREWAY
COVER, AND ASSOCIATED METHOD**

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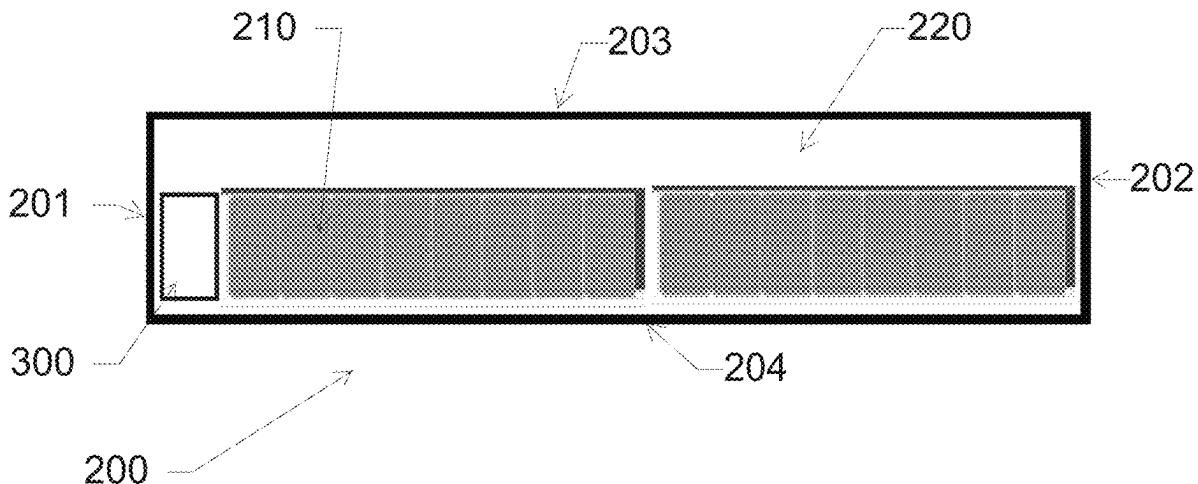
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(57) **ABSTRACT**
A system, including: a roof deck; a first photovoltaic module
above the roof deck, where the first photovoltaic module
includes: a first solar cell, where the first photovoltaic
module includes a first end, a second end opposite the first
end; a first wireway portion above the roof deck, where the
first wireway portion is proximate the first end of the first
photovoltaic module; and a first cover, where the first cover
is removably attached to the first wireway portion, where the
first cover comprises: a second solar cell.

20 Claims, 2 Drawing Sheets



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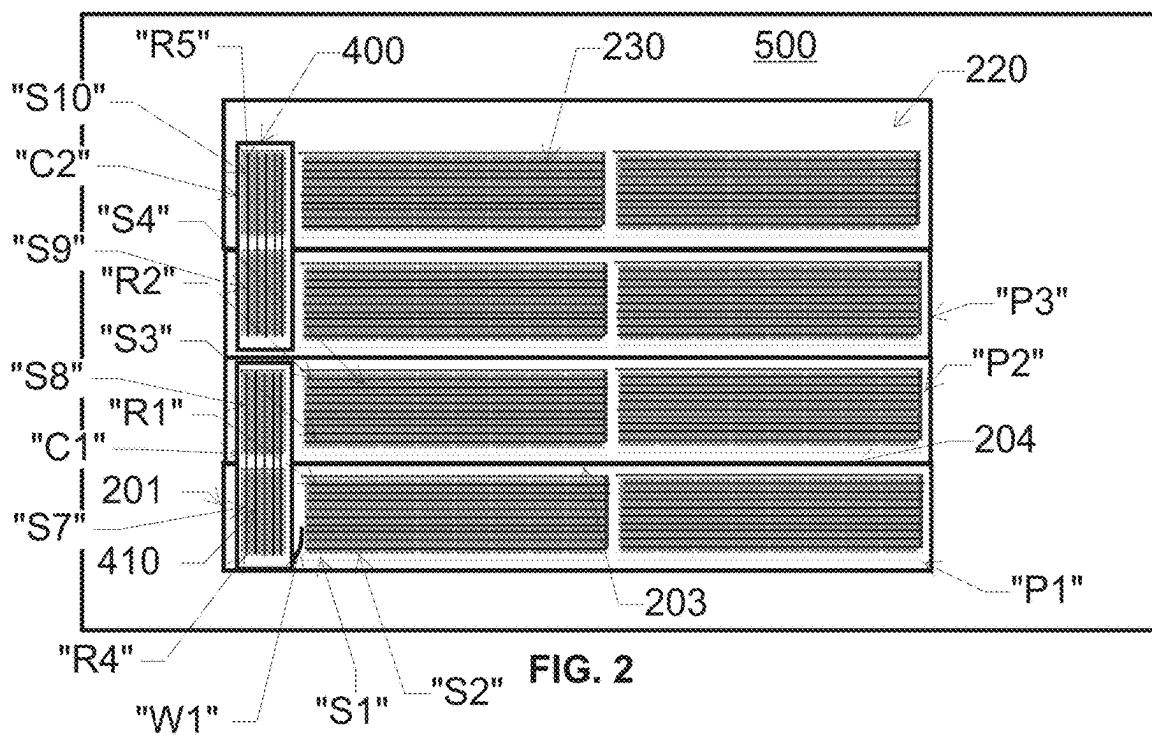
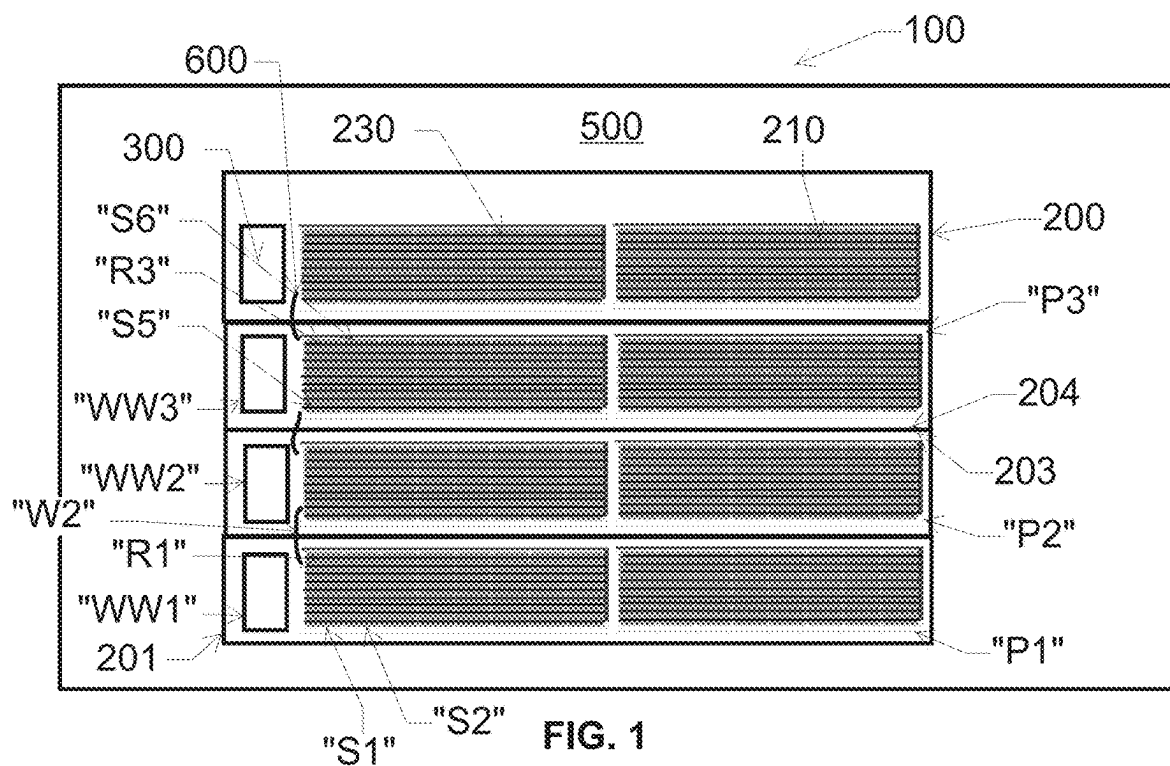
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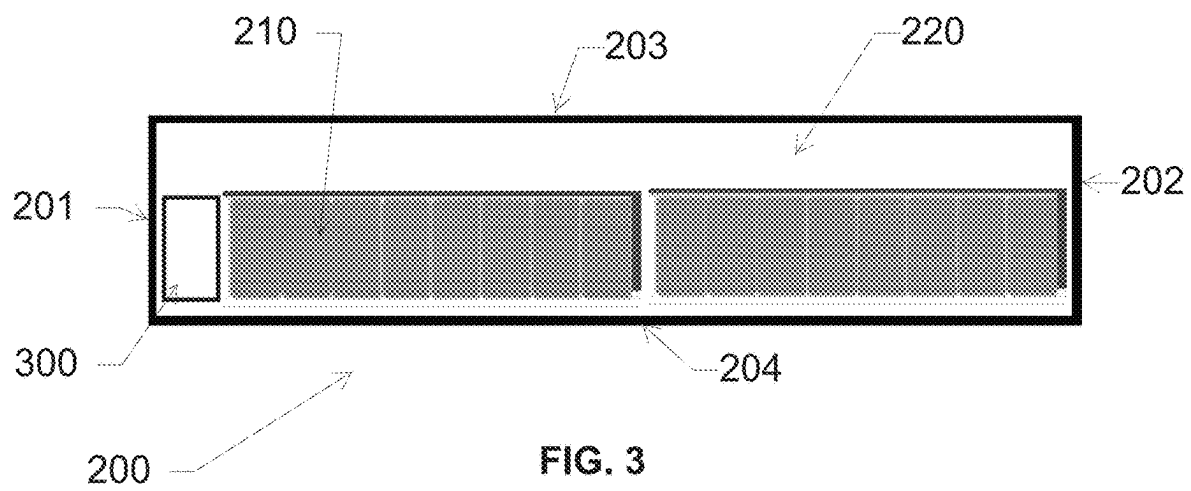
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ROOFING SYSTEM INCLUDING PHOTOVOLTAIC MODULE WIREWAY COVER, AND ASSOCIATED METHOD

CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority to U.S. provisional application No. 63/486,219, titled "ROOFING SYSTEM INCLUDING PHOTOVOLTAIC MODULE WIREWAY COVER, AND ASSOCIATED METHOD," filed Feb. 21, 2023, the disclosure of which is incorporated by reference herein in its entirety.

FIELD

The present invention relates to a roofing system, and, more particularly, to a roofing system that includes a photovoltaic module, a wireway, and a wireway cover.

BACKGROUND

Photovoltaic systems having solar panels are commonly installed on roofing of structures.

SUMMARY

The Claims, rather than the Summary, define covered embodiments of the present invention. The Summary is a high-level overview of various aspects of the invention, and introduces some concepts that are further described in the Detailed Description below. The Summary is not intended to identify key or essential features of the claimed subject matter, and also is not intended to be used in isolation to determine the scope of the claimed subject matter. Instead, the claimed subject matter should be understood by reference to appropriate portions of the Specification and drawings, as well as each claim.

In some embodiments, the present invention provides a system, comprising: a roof deck; a first photovoltaic module above the roof deck, wherein the first photovoltaic module comprises a first solar cell, wherein the first photovoltaic module includes a first end, a second end opposite the first end, a first edge extending between the first end and the second end, and a second edge opposite the first edge, and extending between the first end and the second end; a first wireway portion above the roof deck, wherein the first wireway portion is proximate the first end of the first photovoltaic module; and a first cover, wherein the first cover is removably attached to the first wireway portion, wherein the first cover comprises a second solar cell.

In some embodiments, the first photovoltaic module further comprises: a third solar cell, and a first ribbon, wherein the first solar cell is proximate the first end of the first photovoltaic module, wherein the third solar cell is proximate the first solar cell, wherein the first ribbon extends between the first solar cell and the third solar cell in a first direction, wherein the first ribbon is in electrical communication with the first solar cell and the third solar cell, and wherein the first cover further comprises: a fourth solar cell, and a second ribbon, wherein the second ribbon extends between the second solar cell and the fourth solar cell in a second direction, wherein the second ribbon is in electrical communication with the second solar cell and the fourth solar cell, wherein the second direction is different than the first direction.

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In some embodiments, the second direction is perpendicular to the first direction.

In some embodiments, the system further comprises: a second photovoltaic module above the roof deck, wherein the second photovoltaic module comprises: a third solar cell; a second wireway portion above the roof deck, wherein one side of the second wireway portion is proximate an end of the second photovoltaic module, wherein another side of the second wireway portion is proximate a side of the first wireway portion; and a second cover, wherein the second cover is removably attached to the second wireway portion, wherein the second cover comprises a fourth solar cell.

In some embodiments, the system further comprises: a second photovoltaic module above the roof deck, wherein the second photovoltaic module comprises: a third solar cell, wherein a portion of second photovoltaic module is proximate the second edge of the first photovoltaic module, wherein the first wireway portion is proximate an end of the second photovoltaic module, wherein the first cover further comprises: a fourth solar cell, wherein the fourth solar cell is in electrical communication with the second solar cell.

In some embodiments, the system further comprises: a first wire, wherein the first wire is in electrical communication with the first solar cell of the first photovoltaic module and the second solar cell of the first wireway portion; and a second wire, wherein the second wire is in electrical communication with the third solar cell of the second photovoltaic module and the first solar cell of the first photovoltaic module, and wherein an uppermost end of the ribbon is in electrical communication with the first solar cell of the first photovoltaic module.

In some embodiments, the system further comprises: a second photovoltaic module above the roof deck, wherein the second photovoltaic module comprises: a third solar cell, wherein a bottom edge of the second photovoltaic module is proximate the second edge of the first photovoltaic module; a third photovoltaic module above the roof deck, wherein the third photovoltaic module comprises: a fourth solar cell, wherein a bottom edge of the third photovoltaic module is proximate a top edge of the second photovoltaic module; and a second wireway portion above the roof deck, wherein one side of the second wireway portion is proximate an end of the third photovoltaic module; and wherein the first cover further comprises: a fifth solar cell, and a ribbon, wherein the ribbon extends between the second solar cell and the fifth solar cell, wherein the ribbon is in electrical communication with the second solar cell and the fifth solar cell, wherein the first cover is removably attached to the first and second wireway portions.

In some embodiments, the system further comprises: a first wire, wherein the first wire, wherein the first wire is in electrical communication with the first solar cell of the first photovoltaic module and the second solar cell of the first cover; a second wire, wherein the second wire is in electrical communication with the first solar cell of the first photovoltaic module and the third solar cell of the second photovoltaic module; and a third wire, wherein the third wire is in electrical communication with the fourth solar cell of the third photovoltaic module and the third solar cell of the second photovoltaic module.

In some embodiments, the first cover comprises: a top surface, wherein the top surface comprises a channel, wherein the second solar cell is within the channel.

In some embodiments, the first wireway portion comprises at least one slot, wherein the first cover comprises at least one rail, wherein the at least one rail is removably retained within the at least one slot.

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In some embodiments, the present invention provides a method, comprising: obtaining a first photovoltaic module, wherein the first photovoltaic module comprises: a first solar cell, wherein the first photovoltaic module includes a first end, a second end opposite the first end, a first edge extending between the first end and the second end, and a second edge opposite the first edge, and extending between the first end and the second end; obtaining a first wireway portion; obtaining a first cover, wherein the first cover comprises: a second solar cell; installing the first photovoltaic module above the roof deck; installing the first wireway portion above the roof deck, such that the first wireway portion is proximate the first end of the first photovoltaic module; and removably attaching the first cover to the first wireway portion.

In some embodiments, the method further comprises: obtaining a second photovoltaic module, wherein the second photovoltaic module comprises: a third solar cell; obtaining a second wireway portion; obtaining a second cover, wherein the second cover comprises a fourth solar cell; installing the second photovoltaic module above the roof deck; installing the second wireway portion above the roof deck, such that one side of the second wireway portion is proximate an end of the second photovoltaic module; and removably attaching the second cover to the second wireway portion.

In some embodiments, the method further comprises: obtaining a second photovoltaic module, wherein the second photovoltaic module comprises: a third solar cell; and installing the second photovoltaic module above the roof deck, such that a portion of second photovoltaic module is proximate the second edge of the first photovoltaic module, and an end of the second photovoltaic module is proximate the first wireway portion.

In some embodiments, the first cover further comprises: a fourth solar cell, wherein the fourth solar cell is in electrical communication with the second solar cell; and the method further comprises: obtaining a first wire; obtaining a second wire; installing the first wire to electrically connect the first solar cell of the first photovoltaic module and the second solar cell of the first wireway portion; and installing the second wire to electrically connect the third solar cell of the second photovoltaic module and the first solar cell of the first photovoltaic module.

In some embodiments, the first cover further comprises: a fifth solar cell, and a ribbon, wherein the ribbon extends between the second solar cell and the fifth solar cell, wherein the ribbon is in electrical communication with the second solar cell and the fifth solar cell; wherein the method further comprises: obtaining a second photovoltaic module, wherein the second photovoltaic module comprises: a third solar cell; obtaining a third photovoltaic module, wherein the third photovoltaic module comprises: a fourth solar cell; obtaining a second wireway portion; installing the second photovoltaic module above the roof deck, such that a bottom edge of the second photovoltaic module is proximate the second edge of the first photovoltaic module; installing the third photovoltaic module above the roof deck, such that a bottom edge of the third photovoltaic module is proximate a top edge of the second photovoltaic module; installing the second wireway portion above the roof deck, such that one side of the second wireway portion is proximate an end of the third photovoltaic module, and another side of the second wireway portion is proximate a side of the first wireway portion; and removably attaching the first cover to the first and second wireway portions.

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In some embodiments, the method further comprises: obtaining a first wire; obtaining a second wire; obtaining a third wire; installing the first wire to electrically connect the first solar cell of the first photovoltaic module and the second solar cell of the first cover; installing the second wire to electrically connect the first solar cell of the first photovoltaic module and the third solar cell of the second photovoltaic module; and installing the third wire to electrically connect the fourth solar cell of the third photovoltaic module and the third solar cell of the second photovoltaic module.

In some embodiments, the present invention provides a system, comprising: a roof deck; a first photovoltaic module above the roof deck, wherein the first photovoltaic module comprises a first solar cell, wherein the first photovoltaic module includes a first end, a second end opposite the first end, a first edge extending between the first end and the second end, and a second edge opposite the first edge, and extending between the first end and the second end; a second photovoltaic module proximate the first photovoltaic module above the roof deck, wherein the second photovoltaic module comprises a second solar cell, wherein the second photovoltaic module includes a first end, a second end opposite the first end, a first edge extending between the first end and the second end, and a second edge opposite the first edge, and extending between the first end and the second end; a first wireway portion above the roof deck, wherein the first wireway portion is proximate the first end of the first photovoltaic module; a second wireway portion above the roof deck, wherein the second wireway portion is proximate the first end of the second photovoltaic module; a first cover, wherein the first cover is removably attached to the first wireway portion and the second wireway portion, wherein the first cover comprises a third solar cell and a fourth solar cell; a first wire, wherein the first wire is in electrical communication with the first solar cell of the first photovoltaic module and the second solar cell of the second photovoltaic module; a second wire, wherein the second wire is in electrical communication with the first solar cell of the first photovoltaic module and the third solar cell of the cover; and a ribbon, wherein the ribbon is in electrical communication with the third solar cell of the cover and the fourth solar cell of the first cover, wherein the ribbon provides an electrical connection from a lower end of the first cover to an upper end of the first cover, and wherein the system does not include a return wire electrically connecting the lower end of the first cover to the upper end of the first cover.

BRIEF DESCRIPTION OF THE FIGURES

This section refers to the drawings that form a part of this disclosure, and which illustrate some of the embodiments of structure, materials, and/or methods of the present invention described herein.

FIG. 1 shows a top view of an embodiment of a roofing system, in accordance with some embodiments of the invention;

FIG. 2 shows a top view of an embodiment of a roofing system, in accordance with some embodiments of the invention; and

FIG. 3 shows a top view of an embodiment of a photovoltaic module, in accordance with some embodiments of the invention.

DETAILED DESCRIPTION

In addition to the benefits and improvements that the Specification discloses, other objects and advantages of that

the Specification provides will become apparent from the following description taken in conjunction with the accompanying figures. Although the description discloses and describes detailed embodiments of the present disclosure, the disclosed embodiments are merely illustrative of the disclosure that may be embodied in various forms. In addition, each of the examples given regarding the various embodiments of the disclosure are intended to be illustrative, and not restrictive.

Throughout the specification and claims, the following terms take the meanings explicitly associated herein, unless the context clearly dictates otherwise. The phrases “in an embodiment,” “in some embodiments,” and any similar phrase, as used herein, do not necessarily refer to the same embodiment or embodiments, though the phrases may refer to the same embodiment or embodiments. Furthermore, the phrases “in another embodiment,” and any similar phrase, as used herein, do not necessarily refer to a different embodiment, although the phrases may refer to a different embodiment. All embodiments of the disclosure are intended to be combinable without departing from the scope or spirit of the disclosure.

As used herein, terms such as “comprising,” “including,” “having,” and any similar phrase, do not limit the scope of a specific claim to the materials or steps recited by the claim.

As used herein, a “steep slope roof” is a roof that is disposed on a building, such as a house or other residential structure, having a pitch of Y/X, where Y and X are in a ratio of 4:12 to 20:12, where Y corresponds to the “rise” of the roof, and where X corresponds to the “run” of the roof.

As used herein, an “active” solar cell is a solar cell that converts light (such as, but not limited to, sunlight) into electricity by the photovoltaic effect.

As used herein, an “inactive” solar cell is one that does not convert light into electricity.

This application described components of a roofing system, including, for example a photovoltaic module, a wireway that is made up of wireway portions at ends of the photovoltaic modules, and a cover (also referred to as a wireway cover, and/or lid) on a roof deck. Any or all of these components, and/or other components, described herein may be in accordance with the corresponding description in U.S. Pat. No. 11,404,997, titled “Photovoltaic Shingles and Methods of Installing Same,” issued Aug. 2, 2022; U.S. Pat. No. 11,251,744, titled “Photovoltaic Shingles and Methods of Installing Same,” issued Feb. 15, 2022; and PCT International Patent Publication No. WO 2022/051593, Application No. PCT/US2021/049017, published Mar. 10, 2022, titled “Building Integrated Photovoltaic System,” all of the foregoing owned by GAF Energy LLC, and the disclosures of each of the foregoing patent properties are incorporated by reference herein in their entirety. The roofing system may be installed on a steep slope roof, on a roof that is sloped but has a slope less than that of a steep slope roof, or on a flat roof (that is, a roof with minimal, or no slope).

In some embodiments, the present invention may provide a cover that includes at least one solar cell.

In some embodiments, the solar cell of the wireway cover may be electrically active—that is, the solar cell may convert light into electricity by the photovoltaic effect.

In some embodiments, the wireway portion may be proximate one of more photovoltaic modules on the roof deck. In some embodiments, the wireway portion may include an interior, in which may be disposed one or more components associated with control and/or operation of the photovoltaic modules. In some embodiments, one or more of the wireways may omit any or all of these components, and/or of

another component or components, disposed within its interior. In some embodiments, ends of the wireway portions may be proximate one another, such that the wireway portions form a wireway.

FIGS. 1 and 2 show top views of embodiments of a roofing system installed above a roof deck, where FIG. 2 includes a cover, while FIG. 1 does not include a cover; and FIG. 3 shows a top view of an embodiment of a photovoltaic module, in accordance with some embodiments of the invention.

As the figures illustrate, in some embodiments, a roofing system 100 may include one or more photovoltaic modules 200, one or more wireway portions 300, where one or more sides of the wireway portions 300 are proximate portions of the photovoltaic modules 200, and one or more covers 400, installed above a roof deck 500. As discussed, any or all of the photovoltaic module 200, the wireway portion 300, and the cover 400 may be in accordance with the corresponding description in U.S. Pat. No. 11,404,997, titled “Photovoltaic Shingles and Methods of Installing Same,” issued Aug. 2, 2022; U.S. Pat. No. 11,251,744, titled “Photovoltaic Shingles and Methods of Installing Same,” issued Feb. 15, 2022; and PCT International Patent Publication No. WO 2022/051593, Application No. PCT/US2021/049017, published Mar. 10, 2022, titled “Building Integrated Photovoltaic System,” all of the foregoing owned by GAF Energy LLC, and the disclosures of each of the foregoing patent properties are incorporated by reference herein in their entirety.

Although not shown in the drawings, in some embodiments, the roofing system 100 may include other components, such as but not limited to an underlayment (such as between the photovoltaic modules 200 and the roof deck 500), and/or non-photovoltaic roofing modules (for example, roofing shingles without any solar cell), proximate or spaced apart from the photovoltaic modules 200.

In some embodiments, the roofing system 100 may include one photovoltaic module 200. In some embodiments, the roofing system 100 may include two or more photovoltaic modules 200. In some embodiments, the number of photovoltaic modules 200 in the roofing system 100 may be limited by the area of the roof deck 500 and the area of the photovoltaic module(s) 200.

In some embodiments, any or all of the photovoltaic modules 200 may include at least one solar cell 210. In some embodiments, the at least one solar cell 210 may be above a backsheet 220 of the photovoltaic module. In some embodiments, the backsheet 220 may be a nailable backsheet—that is, the backsheet 220 may be attachable to the roof deck 500 by one or more of the following types of mechanical fasteners that pierce the backsheet 220 and/or the roof deck 500: one or more nails; one or more screws; one or more staples; and/or one or more rivets; and/or combinations or one or more of these fasteners. In some embodiments, the backsheet 220 may not be a nailable backsheet. In some embodiments, whether or not the backsheet 220 is nailable, the backsheet may be attachable to the roof deck 500 by one or more of welding (heat and/or ultrasonic, for example), and/or adhering using an adhesive.

In some embodiments, the at least one solar cell 210 may be attachable to the backsheet 220, and/or to another component or layer of the backsheet 220, by one or more of: a mechanical fastener of fasteners, welding (heat and/or ultrasonic, for example), adhering using an adhesive, and/or laminating.

In some embodiments, the at least one solar cell 210 may be indirectly attached to the backsheet 220—that is, there

may be at least one intervening layer between the at least one solar cell **210** and the backsheet **220**. In some embodiments, there may be one intervening layer between the at least one solar cell **210** and the backsheet **220**. In some embodiments, there may be two or more intervening layers between the at least one solar cell **210** and the backsheet **220**. In some embodiments, there may be at least two intervening layers between the at least one solar cell **210** and the backsheet **220**.

In some embodiments, as shown in the figures, the photovoltaic module **200** may include two sets of the solar cells **210**, where each set of the solar cells **210** includes eight (8) of the solar cells **210**, where the sets of the solar cells **210** are separated from each other by a space. In some embodiments, however, the photovoltaic module **200** may include one set of the solar cells **210**, or more than two sets of the solar cells **210**. In some embodiments, each of the sets of the solar cells **210** may include less than eight of the solar cells **210**, or more than eight of the solar cells **210**. In some embodiments, the sets of the solar cells **210** may have a different number (or the same number) of the solar cells **210** as another set or other sets of the solar cells **210**. In some embodiments, the sets of the solar cells **210** may not be separated from one another by a space.

In some embodiments, the photovoltaic module **200** includes at least one ribbon **230**. In some embodiments, the ribbons **230** may extend between, and/or may electrically connect (that is, may be in electrical communication with), components of the photovoltaic module **200**. In some embodiments, the ribbons **230** may extend between, and/or may electrically connect, the solar cells **210**.

In some embodiments, as shown in the figures, each of the solar cells **210** may be attached to nine (9) of the ribbons **230**. In some embodiments, each of the solar cells **210** may be attached to less than nine of the ribbons **230**, or may be attached to more than nine of the ribbons **230**. In some embodiments, different ones of the solar cells **210** may be attached to a different number of the ribbons **230** than one or more others of the solar cells **210**; or may be attached to the same number of the ribbons **230** as some of the solar cells **210** as well as a different number of the ribbons **230** than others of the solar cells **210**. In some embodiments, the ribbons **230** may connect to, may extend between, and/or may provide an electrical connection with, some of the solar cells **210**.

By way of example, in some embodiments, as shown in the drawings, ribbon “R1” may extend between solar cell “S1” and solar cell “S2” of photovoltaic module “P1,” thereby providing an electrical connection between solar cell “S1” and solar cell “S2.” By way of another example, in some embodiments, as shown in the drawings, ribbon “R2” may extend between solar cell “S3” and solar cell “S4” of photovoltaic module “P2,” thereby providing an electrical connection between solar cell “S3” and solar cell “S4.” By way of another example, in some embodiments, as shown in the drawings, ribbon “R3” may extend between solar cell “S5” and solar cell “S6” of photovoltaic module “P3,” thereby providing an electrical connection between solar cell “S5” and solar cell “S6.”

In some embodiments, each of the photovoltaic modules **200** may have a first end **201** (for example, a left end), a second end **202** (for example, a right end), a first edge **203** (for example, a top edge), and a second edge **204** (for example, a bottom edge). In some embodiments, the first end **201** and the second end **202** may be opposite each other. In some embodiments, the first edge **203** may extend between the first end **201** and the second end **202**. In some embodiments, the first edge **203** and the second edge **204** may be

opposite one another, and the second edge **204** also may extend between the first end **201** and the second end **202**.

In some embodiments, as shown in the drawings, the first edge **203** of photovoltaic module “P1” may be proximate the second edge **204** of photovoltaic module “P2.” Also as shown, in some embodiments, the first edge **203** of photovoltaic module “P2” may be proximate the second edge **204** of photovoltaic module “P3.”

As shown in FIGS. **1** and **2**, in some embodiments, the roofing system **100** may include one or more wireway portions **300**. In some embodiments, the roofing system **100** may include only one wireway portion **300**, and thus the wireway portion **300** may be a wireway. In some embodiments, as the figures show, the roofing system **100** may include two or more wireway portions **300**, wherein one end of one wireway portion **300** is proximate one end of another wireway portion **300**, such that the wireway portions **300** forming the wireway. As discussed, in some embodiments, the wireway portion **300** may include one or more components associated with control and/or operation of the photovoltaic modules **200**, and/or other components of the roofing system **100**.

In some embodiments, as shown in the drawings, the wireway portion **300** may be installed on the photovoltaic module **200**. In some embodiments, the wireway portion **300** may be installed on the backsheet **220** of the photovoltaic module **200**. In some embodiments, the wireway portion **300** may be installed on another layer of the photovoltaic module **200**. In some embodiments, the wireway portion **300** may be installed off of, while proximate, the photovoltaic module **200**. In some embodiments, the wireway portion **300** may be installed off of, as well as spaced apart from, the photovoltaic module **200**.

In some embodiments, the wireway portion **300** may be installed proximate an end portion of the photovoltaic module **200**. Thus, as shown in the figures, in some embodiments, the wireway portion **300** may be installed proximate the first end **201** of the photovoltaic module **200**. In some embodiments, the wireway portion **300** may be installed proximate the second end **202** of the photovoltaic module **200**. In some embodiments, the wireway portion **300** may be installed proximate the first edge **203** of the photovoltaic module **200**. In some embodiments, the wireway portion **300** may be installed proximate the second edge **204** of the photovoltaic module **200**. In some embodiments, the wireway portion **300** may be installed spaced apart from the first end **201**, the second end **202**, the first edge **203**, and/or the second edge **204** of the photovoltaic module **200**.

By way of example, in some embodiments, as shown in the drawings, the first end **201** of photovoltaic module “P1” is proximate wireway portion “WW1.” By way of further example, in some embodiments, the first end **201** of photovoltaic module “P2” is proximate wireway portion “WW2.” By way of further example, the first end **201** of photovoltaic module “P3” is proximate wireway portion “WW3.” By way of further examples, an end of wireway portion “WW1” is proximate an end of wireway portion “WW2,” and an opposite end of wireway portion “WW2” is proximate an end of wireway portion “WW3.”

As discussed, further details of some embodiments of the wireway portions **300** that form the wireway may be in accordance with the corresponding description in U.S. Pat. No. 11,404,997, titled “Photovoltaic Shingles and Methods of Installing Same,” issued Aug. 2, 2022; U.S. Pat. No. 11,251,744, titled “Photovoltaic Shingles and Methods of Installing Same,” issued Feb. 15, 2022; and PCT International Patent Publication No. WO 2022/051593, Application

No. PCT/US2021/049017, published Mar. 10, 2022, titled “Building Integrated Photovoltaic System,” all of the foregoing owned by GAF Energy LLC, and the disclosures of each of the foregoing patent properties are incorporated by reference herein in their entireties.

As shown in FIGS. 1 and 2, in some embodiments, the roofing system 100 may one or more covers 400. In some embodiments, the roofing system 100 may include one cover 400. In some embodiments, the roofing system 100 may include more than one cover 400. In some embodiments, the roofing system 100 may include one cover 400 that is sized (that is, has a length sufficient) to cover all of the wireway portions 300 of the roofing system 100. In some embodiments, the roofing system 100 may include multiple covers 400 to cover all of the wireway portions 300 of the roofing system 100. In some embodiments, the roofing system 100 may include multiple covers 400 of the same length. In some embodiments, the roofing system 100 may include multiple covers 400 of different lengths.

In some embodiments, as shown in the drawings, the cover 400 may be installed on the photovoltaic module 200. In some embodiments, as shown in the drawings, the cover 400 may be installed on the wireway portion 300. In some embodiments, the cover 400 may be removably installed (that is, may be able to be removed without damage to either or both of the cover 400 and/or the wireway portion 300) on the wireway portion 300. In some embodiments, the cover 400 may be permanently or irremovably installed on the wireway portion 300.

In some embodiments, the cover 400 may be installed on one wireway portion 300. In some embodiments, the cover 400 may be installed on two wireway portions 300. In some embodiments, the cover 400 may be installed on more than two wireway portions 300. In some embodiments, a single cover 400 may be installed on each of the wireway portions 300 on the roof deck 500.

By way of example, in some embodiments, as shown in the drawings, cover “C1” is removably attached to wireway portion “WW1” as well as to wireway portion “WW2.” By way of further example, in some embodiments, cover “C2” is removably attached to wireway portion “WW3.” In some embodiments, one cover 400 may attach to one wireway portion 300, two wireway portions 300, three wireway portions 300, or all of the wireway portions 300 of the roofing system 100.

In some embodiments, either the wireway portion 300 or the cover 400 may include a rail, and the other one of the cover 400 and the wireway portion 300 may include a slot. Accordingly, in some embodiments, the rail is received at least partially within the slot, thereby to attach (removably or irremovably) the cover 400 to the wireway portion 300. In some embodiments, other cooperating fasteners may be included in either or both of the wireway portion 300 and the cover 400, thereby to permit removable or irremovable attachment of the cover 400 to the wireway portion 300.

As discussed, further details of some embodiments of the cover 400 may be in accordance with the corresponding description in U.S. Pat. No. 11,404,997, titled “Photovoltaic Shingles and Methods of Installing Same,” issued Aug. 2, 2022; U.S. Pat. No. 11,251,744, titled “Photovoltaic Shingles and Methods of Installing Same,” issued Feb. 15, 2022; and PCT International Patent Publication No. WO 2022/051593, Application No. PCT/US2021/049017, published Mar. 10, 2022, titled “Building Integrated Photovoltaic System,” all of the foregoing owned by GAF Energy

LLC, and the disclosures of each of the foregoing patent properties are incorporated by reference herein in their entireties.

As shown in the drawings, in some embodiments, the cover 400 may include at least one solar cell 410. In some embodiments, the cover 400 may include two solar cells 410. In some embodiments, depending on the length of the cover 400, for example, the cover 400 may include more than two solar cells 410. In some embodiments, the solar cells 410 may be disposed in a same orientation as one or more of the solar cells 210 of the photovoltaic modules 200 in the roofing system 100. In some embodiments, the solar cells 410 may be disposed in a same orientation as all of the solar cells 210 of the photovoltaic modules 200 in the roofing system 100. In some embodiments, the solar cells 410 may be disposed in a different orientation than at least one of the solar cells 210 of the photovoltaic modules 200 in the roofing system 100. In some embodiments, the solar cells may be disposed in a different orientation than all of the solar cells 210 of the photovoltaic modules 200 in the roofing system 100.

In some embodiments, the at least one solar cell 410 may be disposed on a top surface of the cover 400. In some embodiments, at least one solar cell 410 may be disposed within a channel or channels of the cover 400, such as a channel or channels in the top surface. In some embodiments, the one or more solar cells 410 may be secured, bonded, or fastened to the cover 400. In some embodiments, the one or more solar cells 410 may be mechanically fastened to the cover 400, such as by one or more nails, screws, rivets, and/or staples. In some embodiments, the one or more solar cells 410 may be adhered to the cover 400, such as by an adhesive. In some embodiments, the one or more solar cells 410 may be welded to the cover 400, such as by ultrasonic or heat welding. In some embodiments, the one or more solar cells 410 may be retained with the channel or channels of the cover 400 by friction. In some embodiments, the one or more solar cells 410 may be snap fit into the cover 400. In some embodiments, the one or more solar cells 410 may be adhered to a surface of the cover 400. In some embodiments, the one or more solar cells 410 may be removably attached to the cover 400 (that is, removable without damage to the cover 400 and/or the solar cell 410). In some embodiments, the one or more solar cells 410 may be irremovably attached to the cover 400 (that is, may not be removed without damage to the cover 400 and/or the solar cell 410).

As discussed, one or more of the solar cells 410 may be active solar cells—that is, may convert light into electricity. In some embodiments, all of the solar cells 410 of one of the covers 400 are active. In some embodiments, at least one of the solar cells 410 of one of the covers 400 is active, and at least one other solar cell 410 is inactive. In some embodiments, at least one solar cell 410 of the roofing system 100 is active. In some embodiments, all of the solar cells 410 in the roofing system 100 are active. In some embodiments, at least one of the solar cells 410 of the roofing system 100 is active, and at least one other solar cell 410 is inactive.

In some embodiments, each of the solar cells 410 may be attached to at least one ribbon 430. In some embodiments, as shown in the figures, each of the solar cells 410 may be attached to five (5) of the ribbons 430. In some embodiments, each of the solar cells 410 may be attached to less than five of the ribbons 430, or may be attached to more than five of the ribbons 430. In some embodiments, different ones of the solar cells 410 may be attached to a different number of the ribbons 430 than one or more others of the solar cells

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410, or may be attached to the same number of the ribbons 430 as some of the solar cells 410 as well as a different number of the ribbons 430 than others of the solar cells 410. In some embodiments, the ribbons 430 may extend between, and/or may provide an electrical connection between, some of the solar cells 410.

By way of example, in some embodiments, as shown in the drawings, ribbon "R4" may extend between solar cell "S7" and solar cell "S8" of cover "C1," thereby providing an electrical connection between solar cell "S7" and solar cell "S8." By way of another example, in some embodiments, as shown in the drawings, ribbon "R5" may extend between solar cell "S9" and solar cell "S10" of cover "C2," thereby providing an electrical connection between solar cell "S9" and solar cell "S10."

As shown in the drawings, in some embodiments, one or more of ribbon "R1," ribbon "R2," and/or ribbon "R3" of the photovoltaic module 200 may extend in a first direction. In some embodiments, one or more of ribbon "R4," and/or ribbon "R5" of the cover 400 may extend in a second direction. In some embodiments, the first direction may be perpendicular to the second direction. In some embodiments, the first direction may be parallel to the second direction. In some embodiments, the first direction may be neither perpendicular nor parallel to the second direction.

In some embodiments, the roofing system 100 includes the cover 400 with the active solar cell 410 may provide seventeen (17) solar cells per row, as compared to sixteen (16) solar cells that are provided by a roofing system that does not include an active solar cell on a cover. Accordingly, under certain environmental conditions, the roofing system 100 in accordance with the invention may provide 6.25% more power than a roofing system that does not include the active solar cell on a cover. In some embodiments, in the roofing system 100 that include the cover 400 having the active solar cell 410 and a photovoltaic module 200 with fifteen (15) solar cells, the roofing system 100 may provide 6.7% more power than a roofing system that does not include the active solar cell on a cover.

As shown in FIGS. 1 and 2, the roofing system 100 may include at least one wire 600. In some embodiments, the roofing system 100 may include a plurality of wires 600. In some embodiments, the wire 600 may electrically connect components of the roofing system 100. In some embodiments, the wire 600 is an electrical cable. In some embodiments, the wire 600 is a flat ribbon electrical cable. In some embodiments, the wire 600 may electrically connect the solar cell 210 of a first one of the photovoltaic modules 200 and the solar cell 410 of a first one of the covers 400, such as through a bus or bussing, a wire or wires, and/or one or more additional and/or other electrical components. In some embodiments, the wire 600 may electrically connect the solar cell 210 of the first one of the photovoltaic modules 200 with the solar cell 210 of another, second one of the photovoltaic modules 200. In some embodiments, the wires 600 may be disposed within an interior of the wireway portions 300, and/or otherwise may be covered by the covers 400. In some embodiments, the wire 600 may be electrically connected to one or more bus bars and/or another component or other components of the photovoltaic module 200 and/or the cover 400, to electrically connect one or more of the solar cells 210 and/or one or more of the solar cells 410, for example.

By way of example, in some embodiments, as shown in the drawings, wire "W1" provides an electrical connection between solar cell "S1" of photovoltaic module "P1" and solar cell "S7" of cover "C1." Thus, in some embodiments,

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the solar cells and the ribbons of the cover or covers 400 provide an electrical connection to the solar cells of the photovoltaic module "P1," such that, for example, ends of the ribbons on an uppermost end of the roof deck 500 may provide an electrical connection to the solar cells of the photovoltaic module on a lowermost end on the roof deck 500. Therefore, in some embodiments of the invention, the roofing system 100 may not use a wire, such as a return wire within the wireway made up of the wireway portions 300, to provide an electrical connection at an uppermost end of the wireway on the roof deck, to the photovoltaic module at a lowermost end of the roof deck 500 including an electrical connection to the solar cells thereof. In some embodiments, the roofing system 100 may include a power optimizer, in order to integrate power produced by one or more of the covers 400 including the solar cells 410, with power produced by the solar cells 210 of the photovoltaic modules 200.

Although the figures show the covers 400 with the solar cells 410 spanning two photovoltaic modules 200, one or more of the covers may span one photovoltaic module, and/or may span more than two photovoltaic modules, such as three, four, five, or more than five photovoltaic modules, and in some embodiments may span all of the photovoltaic modules in the column of photovoltaic modules installed on the roof deck 500.

By way of another example, in some embodiments, as shown in the drawings, wire "W2" provides an electrical connection between solar cell "S3" of photovoltaic module "P2" and solar cell "S1" of photovoltaic module "P1."

"In some embodiments, a method comprises obtaining one or more of the components discussed herein, and/or one or more other components, and installing any or all of the components above a roof deck, as described.

Variations, modifications and alterations to embodiments of the present disclosure described above will make themselves apparent to those skilled in the art. All such variations, modifications, alterations and the like are intended to fall within the spirit and scope of the present disclosure, limited solely by the appended claims.

While several embodiments of the present disclosure have been described, it is understood that these embodiments are illustrative only, and not restrictive, and that many modifications may become apparent to those of ordinary skill in the art. For example, all dimensions discussed herein are provided as examples only, and are intended to be illustrative and not restrictive.

Any feature or element that is positively identified in this description may also be specifically excluded as a feature or element of an embodiment of the present as defined in the claims.

The disclosure described herein may be practiced in the absence of any element or elements, limitation or limitations, which is not specifically disclosed herein. Thus, for example, in each instance herein, any of the terms "comprising," "consisting essentially of" and "consisting of" may be replaced with either of the other two terms, without altering their respective meanings as defined herein. The terms and expressions which have been employed are used as terms of description and not of limitation, and there is no intention in the use of such terms and expressions of excluding any equivalents of the features shown and described or portions thereof, but it is recognized that various modifications are possible within the scope of the disclosure.

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The invention claimed is:

1. A system, comprising:
 - a roof deck;
 - a first photovoltaic module installed on the roof deck, wherein the first photovoltaic module comprises a first solar cell;
 - a first wireway portion installed on the roof deck, wherein the first wireway portion is proximate to an end of the first photovoltaic module, wherein the first wireway portion comprises at least one slot; and
 - a first cover, wherein the first cover is removably attached to the first wireway portion, wherein the first cover comprises a second solar cell and at least one rail, and wherein the at least one rail is removably retained within the at least one slot.
2. The system of claim 1, wherein the first photovoltaic module further comprises:
 - a third solar cell, and
 - a first ribbon, wherein the first ribbon extends between the first solar cell and the third solar cell in a first direction, wherein the first ribbon is in electrical communication with the first solar cell and the third solar cell; wherein the first cover further comprises:
 - a fourth solar cell, and
 - a second ribbon, wherein the second ribbon extends between the second solar cell and the fourth solar cell in a second direction, wherein the second ribbon is in electrical communication with the second solar cell and the fourth solar cell, wherein the second direction is different than the first direction.
3. The system of claim 2, wherein the second direction is perpendicular to the first direction.
4. The system of claim 1, further comprising:
 - a second photovoltaic module installed on the roof deck, wherein the second photovoltaic module comprises a third solar cell;
 - a second wireway portion installed on the roof deck, wherein one side of the second wireway portion is proximate the second photovoltaic module, wherein another side of the second wireway portion is proximate the first wireway portion; and
 - a second cover, wherein the second cover is removably attached to the second wireway portion, wherein the second cover comprises a fourth solar cell.
5. The system of claim 1, further comprising:
 - a second photovoltaic module installed on the roof deck, wherein the second photovoltaic module comprises a third solar cell, wherein a portion of second photovoltaic module is proximate the first photovoltaic module, wherein the first wireway portion is proximate the second photovoltaic module; wherein the first cover further comprises a fourth solar cell.

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6. The system of claim 5, further comprising:

- a first wire, wherein the first wire is in electrical communication with the first solar cell of the first photovoltaic module and the second solar cell of the first cover; and
- a second wire, wherein the second wire is in electrical communication with the third solar cell of the second photovoltaic module and the first solar cell of the first photovoltaic module.

7. The system of claim 1, further comprising:

- a second photovoltaic module installed on the roof deck, wherein the second photovoltaic module comprises a third solar cell, wherein the second photovoltaic module is proximate the first photovoltaic module;
- a third photovoltaic module installed on the roof deck, wherein the third photovoltaic module comprises a fourth solar cell, wherein the third photovoltaic module is proximate the second photovoltaic module; and
- a second wireway portion installed on the roof deck, wherein the second wireway portion is proximate the third photovoltaic module,

wherein the first cover further comprises:

- a fifth solar cell, and
- a ribbon, wherein the ribbon extends between the second solar cell and the fifth solar cell, wherein the ribbon is in electrical communication with the second solar cell and the fifth solar cell, wherein the first cover is removably attached to the first wireway portion and the second wireway portion.

8. The system of claim 7, further comprising:

- a first wire, wherein the first wire is in electrical communication with the first solar cell of the first photovoltaic module and the second solar cell of the first cover;
- a second wire, wherein the second wire is in electrical communication with the first solar cell of the first photovoltaic module and the third solar cell of the second photovoltaic module; and
- a third wire, wherein the third wire is in electrical communication with the fourth solar cell of the third photovoltaic module and the third solar cell of the second photovoltaic module, and wherein an end of the ribbon is in electrical communication with the first solar cell of the first photovoltaic module.

9. The system of claim 1, wherein the first cover comprises a top surface, wherein the top surface comprises a channel, wherein the second solar cell is positioned within the channel.

10. A method, comprising:

- obtaining a first photovoltaic module, wherein the first photovoltaic module comprises a first solar cell;
- obtaining a first wireway portion, wherein the first wireway portion comprises at least one slot;
- obtaining a first cover, wherein the first cover comprises a second solar cell and at least one rail;
- installing the first photovoltaic module on a roof deck;

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installing the first wireway portion on the roof deck, such that the first wireway portion is proximate an end of the first photovoltaic module; and
removably attaching the first cover to the first wireway portion,
wherein the at least one rail is removably retained within the at least one slot.

11. The method of claim 10,

wherein the first photovoltaic module further comprises:
a third solar cell, and
a first ribbon,

wherein the first ribbon extends between the first solar cell and the third solar cell in a first direction,
wherein the first ribbon is in electrical communication with the first solar cell and the third solar cell;

wherein the first cover further comprises:

a fourth solar cell, and

a second ribbon,

wherein the second ribbon extends between the second solar cell and the fourth solar cell in a second direction,

wherein the second ribbon is in electrical communication with the second solar cell and the fourth solar cell,

wherein the second direction is different than the first direction.

12. The method of claim 11, wherein the second direction is perpendicular to the first direction.

13. The method of claim 10, further comprising:

obtaining a second photovoltaic module,

wherein the second photovoltaic module comprises a third solar cell;

obtaining a second wireway portion;

obtaining a second cover,

wherein the second cover comprises a fourth solar cell,

wherein the fourth solar cell is in electrical communication with the second solar cell;

installing the second photovoltaic module on the roof deck;

installing the second wireway portion on the roof deck, such that:

one side of the second wireway portion is proximate an end of the second photovoltaic module, and
another side of the second wireway portion is proximate a side of the first wireway portion; and

removably attaching the second cover to the second wireway portion.

14. The method of claim 10, further comprising:

obtaining a second photovoltaic module,

wherein the second photovoltaic module comprises a third solar cell; and installing the second photovoltaic module on the roof deck, such that:

a portion of second photovoltaic module is proximate the first photovoltaic module, and
an end of the second photovoltaic module is proximate the first wireway portion.

15. The method of claim 14, wherein the first cover further comprises a fourth solar cell;

wherein the method further comprises:

obtaining a first wire;

obtaining a second wire;

installing the first wire to electrically connect the first solar cell and the second solar cell; and

installing the second wire to electrically connect the third solar cell and the fourth solar cell.

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16. The method of claim 10, wherein the first cover further comprises:

a fifth solar cell, and

a ribbon,

wherein the ribbon extends between the second solar cell and the fifth solar cell,

wherein the ribbon is in electrical communication with the second solar cell and the fifth solar cell;

wherein the method further comprises:

obtaining a second photovoltaic module,

wherein the second photovoltaic module comprises a third solar cell;

obtaining a third photovoltaic module,

wherein the third photovoltaic module comprises a fourth solar cell;

obtaining a second wireway portion;

installing the second photovoltaic module on the roof deck, such that the second photovoltaic module is proximate the first photovoltaic module;

installing the third photovoltaic module on the roof deck, such that the third photovoltaic module is proximate the second photovoltaic module;

installing the second wireway portion on the roof deck, such that one side of the second wireway portion is proximate an end of the third photovoltaic module; and

removably attaching the first cover is removably attached to the first wireway portion and the second wireway portion.

17. The method of claim 16, further comprising:

obtaining a first wire;

obtaining a second wire;

obtaining a third wire;

installing the first wire to electrically connect the first solar cell and the second solar cell;

installing the second wire to electrically connect the first solar cell and the third solar cell; and

installing the third wire to electrically connect the fourth solar cell and the third solar cell.

18. The method of claim 10, wherein the first cover comprises a top surface, wherein the top surface comprises a channel, wherein the second solar cell is positioned within the channel.

19. A system, comprising:

a roof deck;

a first photovoltaic module installed on the roof deck,

wherein the first photovoltaic module comprises

a first solar cell,

a second solar cell, and

a first ribbon,

wherein the first ribbon extends between the first solar cell and the second solar cell in a first direction,

wherein the first ribbon is in electrical communication with the first solar cell and the second solar cell;

a first wireway portion installed on the roof deck,

wherein the first wireway portion is proximate to an end of the first photovoltaic module; and

a first cover,

wherein the first cover is removably attached to the first wireway portion,

wherein the first cover comprises

a third solar cell,

a fourth solar cell, and

a second ribbon,

wherein the second ribbon extends between the third solar cell and the fourth solar cell in a second direction,

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wherein the second direction is different than the first direction, and

wherein the second ribbon is in electrical communication with the third solar cell and the fourth solar cell.

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20. The system of claim **19**, wherein the second direction is perpendicular to the first direction.

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